

9. Phylogenetic Diversity - Communities

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OVERVIEW

Complementing taxonomic measures of α - and β -diversity with evolutionary information yields insight into a broad range of biodiversity issues including conservation, biogeography, and community assembly. In this worksheet, you will be introduced to some commonly used methods in phylogenetic community ecology.

After completing this assignment you will know how to:

1. incorporate an evolutionary perspective into your understanding of community ecology
2. quantify and interpret phylogenetic α - and β -diversity
3. evaluate the contribution of phylogeny to spatial patterns of biodiversity

Directions:

1. In the Markdown version of this document in your cloned repo, change “Student Name” on line 3 (above) with your name.
2. Complete as much of the worksheet as possible during class.
3. Use the handout as a guide; it contains a more complete description of data sets along with examples of proper scripting needed to carry out the exercises.
4. Answer questions in the worksheet. Space for your answers is provided in this document and is indicated by the “>” character. If you need a second paragraph be sure to start the first line with “>”. You should notice that the answer is highlighted in green by RStudio (color may vary if you changed the editor theme).
5. Before you leave the classroom today, it is *imperative* that you **push** this file to your GitHub repo, at whatever stage you are. This will enable you to pull your work onto your own computer.
6. When you have completed the worksheet, **Knit** the text and code into a single PDF file by pressing the **Knit** button in the RStudio scripting panel. This will save the PDF output in your ‘9.PhyloCom’ folder.
7. After Knitting, please submit the worksheet by making a **push** to your GitHub repo and then create a **pull request** via GitHub. Your pull request should include this file *9.PhyloCom_Worksheet.Rmd* and the PDF output of **Knitr** (*9.PhyloCom_Worksheet.pdf*).

The completed exercise is due on **Wednesday, March 5th, 2025 before 12:00 PM (noon)**.

1) SETUP

Typically, the first thing you will do in either an R script or an RMarkdown file is setup your environment. This includes things such as setting the working directory and loading any packages that you will need.

In the R code chunk below, provide the code to:

1. clear your R environment,

2. print your current working directory,
3. set your working directory to your `Week7-PhyloCom/` folder,
4. load all of the required R packages (be sure to install if needed), and
5. load the required R source file.

```
rm(list = ls())
getwd()
```

```
## [1] "C:/github/QB2026_Stancil/QB2026/Week9-PhyloCom"
```

```
package.list <- c('picante', 'ape', 'seqinr', 'vegan', 'fossil',
                  'reshape', 'devtools', 'BiocManager', 'ineq',
                  'labdsv', 'matrixStats', 'pROC')
#not working of course!
#for (package in package.list) {
#  if (!require(package, character.only = TRUE, quietly = TRUE)) {
#    install.packages(package, repos='http://cran.us.r-project.org')
#    library(package, character.only = TRUE)
#  }
#}

source("../bin/MothurTools.R")
```

```
## Loading required package: reshape
```

```
## Warning: package 'reshape' was built under R version 4.5.2
```

2) DESCRIPTION OF DATA

need to discuss data set from spatial ecology!

We sampled >50 forested ponds in Brown County State Park, Yellowwood State Park, and Hoosier National Forest in southern Indiana. In addition to measuring a suite of geographic and environmental variables, we characterized the diversity of bacteria in the ponds using molecular-based approaches. Specifically, we amplified the 16S rRNA gene (i.e., the DNA sequence) and 16S rRNA transcripts (i.e., the RNA transcript of the gene) of bacteria. We used a program called `mothur` to quality-trim our data set and assign sequences to operational taxonomic units (OTUs), which resulted in a site-by-OTU matrix.

In this module we will focus on taxa that were present (i.e., DNA), but there will be a few steps where we need to parse out the transcript (i.e., RNA) samples. See the handout for a further description of this week's dataset.

3) LOAD THE DATA

In the R code chunk below, do the following:

1. load the environmental data for the Brown County ponds (`20130801_PondDataMod.csv`),
2. load the site-by-species matrix using the `read.otu()` function,
3. subset the data to include only DNA-based identifications of bacteria,
4. rename the sites by removing extra characters,
5. remove unnecessary OTUs in the site-by-species, and
6. load the taxonomic data using the `read.tax()` function from the source-code file.

```

env <- read.table("./data/20130801_PondDataMod.csv", sep = ",", header = TRUE)
env <- na.omit(env)

# Load Site-by-Species Matrix
comm <- read.otu(shared = "./data/INPonds.final.rdp.shared", cutoff = "1")

# Select DNA data using `grep()`
comm <- comm[grep("*-DNA", rownames(comm)), ]

# Perform replacement of all matches with `gsub()`
rownames(comm) <- gsub("\\-DNA", "", rownames(comm))
rownames(comm) <- gsub("\\_", "", rownames(comm))

tax <- read.tax(taxonomy = "./data/INPonds.final.rdp.1.cons.taxonomy")

## Warning in type.convert.default(as.character(x)): 'as.is' should be specified
## by the caller; using TRUE
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified
## by the caller; using TRUE
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified
## by the caller; using TRUE
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified
## by the caller; using TRUE
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified
## by the caller; using TRUE
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified
## by the caller; using TRUE

```

Next, in the R code chunk below, do the following:

1. load the FASTA alignment for the bacterial operational taxonomic units (OTUs),
2. rename the OTUs by removing everything before the tab (\t) and after the bar (|),
3. import the *Methanosarcina* outgroup FASTA file,
4. convert both FASTA files into the DNABin format and combine using `rbind()`,
5. visualize the sequence alignment,
6. using the alignment (with outgroup), pick a DNA substitution model, and create a phylogenetic distance matrix,
7. using the distance matrix above, make a neighbor joining tree,
8. remove any tips (OTUs) that are not in the community data set,
9. plot the rooted tree.

```

# Remove sites not in the environmental data set
comm <- comm[rownames(comm) %in% env$Sample_ID, ]

# Remove zero-abundance OTUs from data set
comm <- comm[ , colSums(comm) > 0]

library(seqinr)

```

```
## Warning: package 'seqinr' was built under R version 4.5.2
```

```
tax <- read.tax(taxonomy = "../data/INPonds.final.rdp.1.cons.taxonomy")
```

```
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified  
## by the caller; using TRUE
```

```
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified  
## by the caller; using TRUE  
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified  
## by the caller; using TRUE  
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified  
## by the caller; using TRUE  
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified  
## by the caller; using TRUE  
## Warning in type.convert.default(as.character(x)): 'as.is' should be specified  
## by the caller; using TRUE
```

```
# Import the alignment file (`seqinr`)  
ponds.cons <- read.alignment(file = "../data/INPonds.final.rdp.1.rep.fasta",  
                             format = "fasta")
```

```
# Rename OTUs in the FASTA File
```

```
# 2023, there was an issue with original gsub code,  
# perhaps due to UTF8 ASCII character issue  
# ponds.cons$nam <- gsub("\\|.*$", "", gsub("^.*?\\t", "", ponds.cons$nam))  
# new gsub() code seems to help:
```

```
ponds.cons$nam <- gsub(".*\\t", "", ponds.cons$nam)  
ponds.cons$nam <- gsub("\\|.*", "", ponds.cons$nam)
```

```
library(ape)
```

```
## Warning: package 'ape' was built under R version 4.5.2
```

```
##  
## Attaching package: 'ape'
```

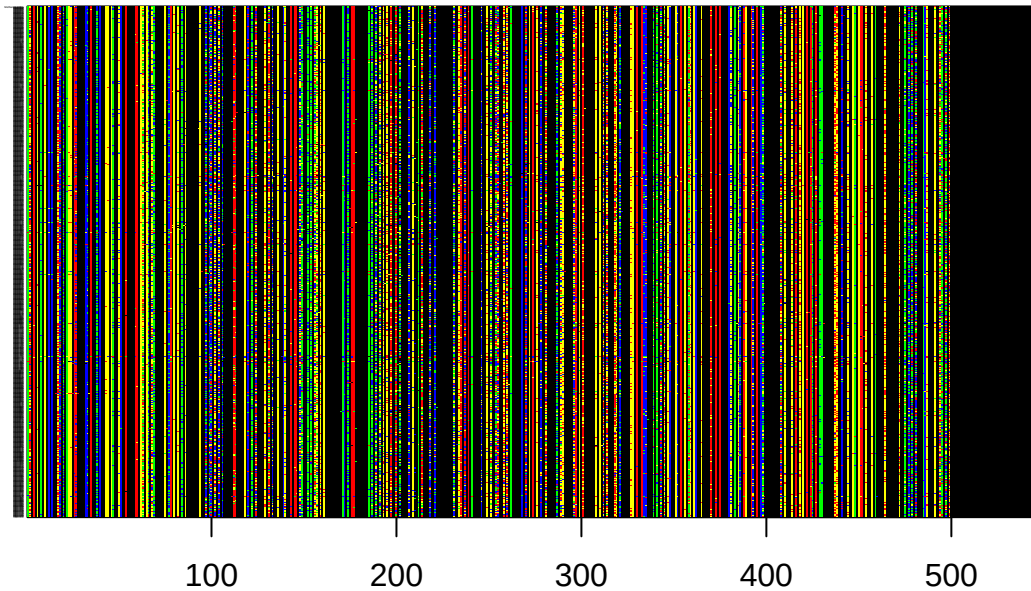
```
## The following objects are masked from 'package:seqinr':  
##  
## as.alignment, consensus
```

```
# Import outgroup sequence  
outgroup <- read.alignment(file = "../data/methanosarcina.fasta", format = "fasta")
```

```
# Convert alignment file to DNABin  
DNABin <- rbind(as.DNABin(outgroup), as.DNABin(ponds.cons))
```

```
# Visualize alignment  
image.DNABin(DNABin, show.labels = T, cex.lab = 0.05, las = 1)
```

■ A
 ■ G
 ■ C
 ■ T
 ■ N
 ■ -



```

# Make distance matrix (`ape`)
seq.dist.jc <- dist.dna(DNAbin, model = "JC", pairwise.deletion = FALSE)

# Make a neighbor-joining tree file (`ape`)
phy.all <- bionj(seq.dist.jc)

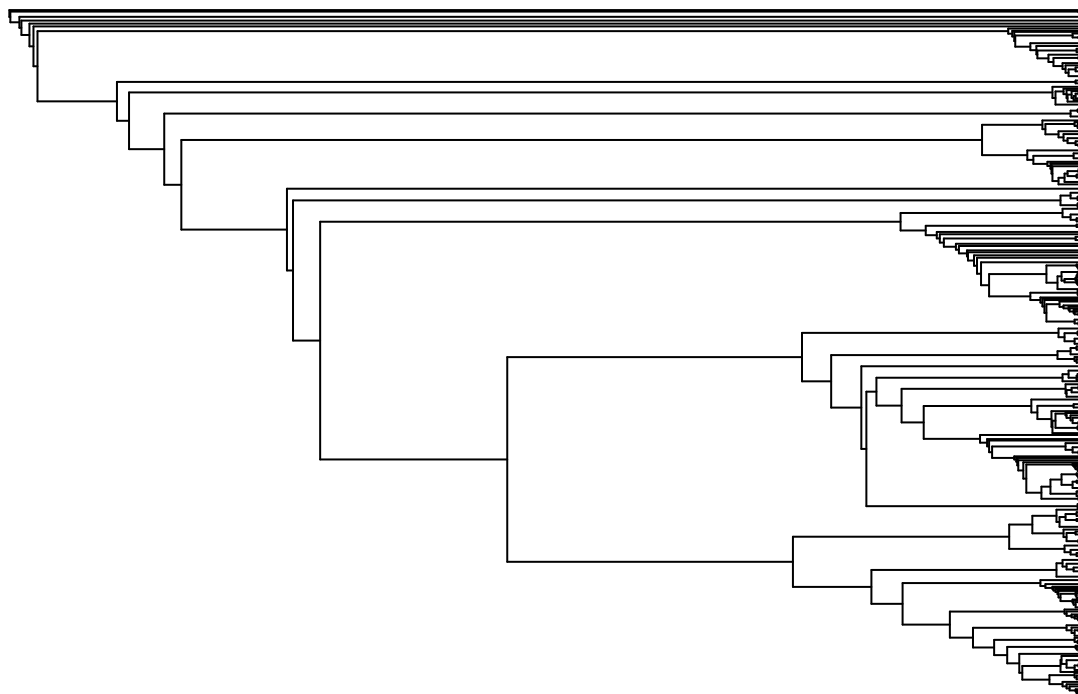
# Drop tips of zero-occurrence OTUs (`ape`)
phy <- drop.tip(phy.all, phy.all$tip.label[!phy.all$tip.label %in%
    c(colnames(comm), "Methanosarcina")])

# Identify outgroup sequence
outgroup <- match("Methanosarcina", phy$tip.label)

# Root the tree {ape}
phy <- root(phy, outgroup, resolve.root = TRUE)

# Plot the rooted tree {ape}
par(mar = c(1, 1, 2, 1) + 0.1)
plot.phylo(phy, main = "Neighbor Joining Tree", "phylogram",
    show.tip.label = FALSE, use.edge.length = FALSE,
    direction = "right", cex = 0.6, label.offset = 1)
    
```

Neighbor Joining Tree



4) PHYLOGENETIC ALPHA DIVERSITY

A. Faith's Phylogenetic Diversity (PD)

In the R code chunk below, do the following:
1. calculate Faith's D using the `pd()` function.

```
library(picante)
```

```
## Warning: package 'picante' was built under R version 4.5.2
```

```
## Loading required package: vegan
```

```
## Warning: package 'vegan' was built under R version 4.5.2
```

```
## Loading required package: permute
```

```
## Warning: package 'permute' was built under R version 4.5.2
```

```
##
```

```
## Attaching package: 'permute'
```

```
## The following object is masked from 'package:seqinr':
##
##      getType
```

```
## Loading required package: nlme
```

```
##
## Attaching package: 'nlme'
```

```
## The following object is masked from 'package:seqinr':
##
##      gls
```

```
pd <- pd(comm, phy, include.root = FALSE)
```

In the R code chunk below, do the following:

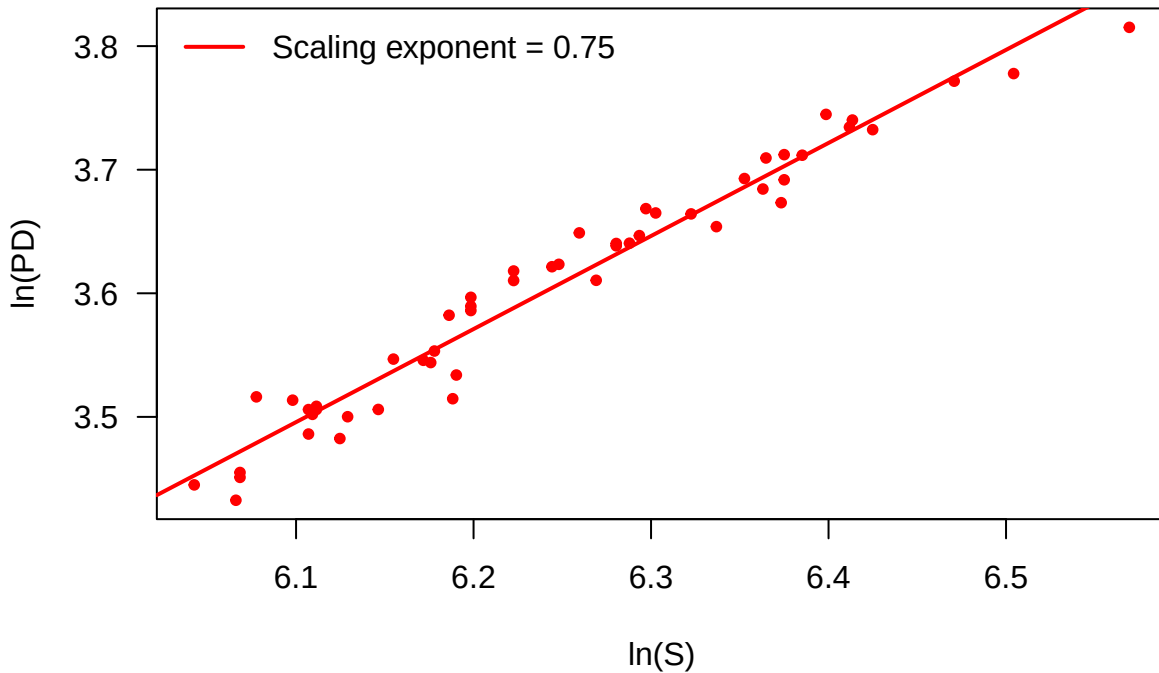
1. plot species richness (S) versus phylogenetic diversity (PD),
2. add the trend line, and
3. calculate the scaling exponent.

```
''' r
# Biplot of S and PD
par(mar = c(5, 5, 4, 1) + 0.1)

plot(log(pd$S), log(pd$PD),
     pch = 20, col = "red", las = 1,
     xlab = "ln(S)", ylab = "ln(PD)", cex.main = 1,
     main="Phylodiversity (PD) vs. Taxonomic richness (S)")

# Test of power-law relationship
fit <- lm('log(pd$PD) ~ log(pd$S)')
abline(fit, col = "red", lw = 2)
exponent <- round(coefficients(fit)[2], 2)
legend("topleft", legend=paste("Scaling exponent = ", exponent, sep = ""),
      bty = "n", lw = 2, col = "red")
```

Phylogenetic diversity (PD) vs. Taxonomic richness (S)



Question 1: Answer the following questions about the PD-S pattern.

a. Based on how PD is calculated, how and why should this metric be related to taxonomic richness? b. When would you expect these two estimates of diversity to deviate from one another? c. Interpret the significance of the scaling PD-S scaling exponent.

Answer 1a: Because Faith's PD is based on evolutionary history shared between extant species, increasing species richness generally increases Faith's PD. **Answer 1b:** Estimates of diversity may differ when communities differ in evolutionary breadth (ecological niche breadth?), meaning a community with species from the same genus will have a low FPD, while a community with species from distinct genera should have higher FPD. **Answer 1c:** The scaling component of 0.75 indicates diminishing returns. Therefore each additional species adds less evolutionary history than the previous one, species are phylogenetically clustered, and communities are dominated by a few clades.

i. Randomizations and Null Models

In the R code chunk below, do the following:

1. estimate the standardized effect size of PD using the `richness` randomization method.

```
ses.pd <- ses.pd(comm[1:2,], phy, null.model = "richness", runs = 25,
                 include.root = FALSE)
ses.pd
```

```
##      ntaxa  pd.obs pd.rand.mean pd.rand.sd pd.obs.rank  pd.obs.z  pd.obs.p
## BC001   668 43.71912   44.03693  0.6427515      10 -0.4944495 0.3846154
## BC002   587 40.94334   39.80322  0.8497665      22  1.3416881 0.8461538
```



```
##      runs
## BC001  25
## BC002  25
```

Question 2: Using `help()` and the table above, run the `ses.pd()` function using two other null models and answer the following questions:

- What are the null and alternative hypotheses you are testing via randomization when calculating `ses.pd`?
- How did your choice of null model influence your observed `ses.pd` values? Explain why this choice affected or did not affect the output.

Answer 2a: H0 : the observed PD should fall within the distribution of PD values generated by random communities of the same richness. HA: communities are not random draws from the phylogeny and they show non-random phylogenetic structure. **Answer 2b:** The richness null model fixes species richness in every random community, therefore SES.PD reflects whether the observed PD is unusually high or low given the richness. Because richness is controlled, the SES.PD values are near 0, meaning the null model removed the strong PD vs. species richness correlation and isolated true phylogenetic structure.

B. Phylogenetic Dispersion Within a Sample

Another way to assess phylogenetic α -diversity is to look at dispersion within a sample.

i. Phylogenetic Resemblance Matrix

In the R code chunk below, do the following:

- calculate the phylogenetic resemblance matrix for taxa in the Indiana ponds data set.

```
phydist <- cophenetic.phylo(phy)
#phydist
```

ii. Net Relatedness Index (NRI)

In the R code chunk below, do the following:

- Calculate the NRI for each site in the Indiana ponds data set.

```
# Estimate standardized effect size of NRI via randomization (`picante`)
ses.mpd <- ses.mpd(comm, phydist, null.model = "taxa.labels",
  abundance.weighted = FALSE, runs = 25)

# Calculate NRI
NRI <- as.matrix(-1 * ((ses.mpd[,2] - ses.mpd[,3]) / ses.mpd[,4]))
rownames(NRI) <- row.names(ses.mpd)
colnames(NRI) <- "NRI"
```

iii. Nearest Taxon Index (NTI)

In the R code chunk below, do the following: 1. Calculate the NTI for each site in the Indiana ponds data set.

```
# Estimate Standardized Effect Size of NRI via Randomization {picante}
ses.mntd <- ses.mntd(comm, phydist, null.model = "taxa.labels",
                    abundance.weighted = FALSE, runs = 25)

# Calculate NTI
NTI <- as.matrix(-1 * ((ses.mntd[,2] - ses.mntd[,3]) / ses.mntd[,4]))
rownames(NTI) <- row.names(ses.mntd)
colnames(NTI) <- "NTI"
```

Question 3:

- In your own words describe what you are doing when you calculate the NRI.
- In your own words describe what you are doing when you calculate the NTI.
- Interpret the NRI and NTI values you observed for this dataset.
- In the NRI and NTI examples above, the arguments “abundance.weighted = FALSE” means that the indices were calculated using presence-absence data. Modify and rerun the code so that NRI and NTI are calculated using abundance data. How does this affect the interpretation of NRI and NTI?

Answer 3a: NRI is based on mean pairwise phylogenetic distances, comparing average phylogenetic distance between all pairs of species in a community. Because this metric considers distance across the entire phylogenetic tree, it captures patterns like clustering and overdispersion. **Answer 3b:** NTI is based on mean nearest taxon distance, calculating average distance to each species closest relative. **Answer 3c:** Based on the negative NRI and NTI values, species are less closely related than expected with the null model. There may be overdispersion of competitive exclusion occurring in this community.

Answer 3d: Done.

5) PHYLOGENETIC BETA DIVERSITY

A. Phylogenetically Based Community Resemblance Matrix

In the R code chunk below, do the following:

- calculate the phylogenetically based community resemblance matrix using Mean Pair Distance, and
- calculate the phylogenetically based community resemblance matrix using UniFrac distance.

```
# Mean Pairwise Distance
dist.mp <- comdist(comm, phydist)
```

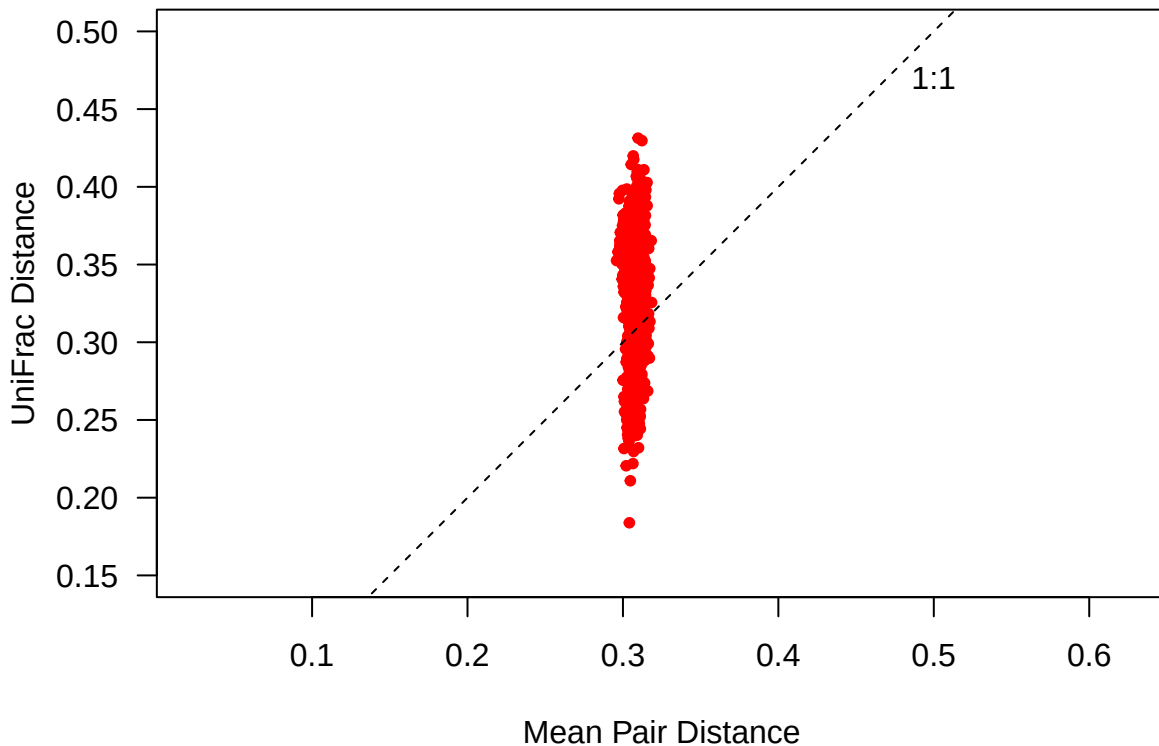
```
## [1] "Dropping taxa from the distance matrix because they are not present in the community data:"
## [1] "Methanosarcina"
```

```
# UniFrac Distance (Note: this takes a few minutes; be patient)
dist.uf <- unifrac(comm, phy)
```

In the R code chunk below, do the following:

- plot Mean Pair Distance versus UniFrac distance and compare.

```
par(mar = c(5, 5, 2, 1) + 0.1)
plot(dist.mp, dist.uf,
     pch = 20, col = "red", las = 1, asp = 1, xlim = c(0.15, 0.5), ylim = c(0.15, 0.5),
     xlab = "Mean Pair Distance", ylab = "UniFrac Distance")
abline(b = 1, a = 0, lty = 2)
text(0.5, 0.47, "1:1")
```



Question 4:

- In your own words describe Mean Pair Distance, UniFrac distance, and the difference between them.
- Using the plot above, describe the relationship between Mean Pair Distance and UniFrac distance.
Note: we are calculating unweighted phylogenetic distances (similar to incidence based measures). That means that we are not taking into account the abundance of each taxon in each site.
- Why might MPD show less variation than UniFrac?

Answer 4a: MPD measures how phylogenetically different two communities are by taking the average of all pairwise distances between species in Comm A and species in Comm B. Unweighted UniFrac measures the fraction of the phylogenetic tree that is unique to each community. MPD calculates the average phylogenetic distance and UniFrac calculates unique branch distance.

Answer 4b: In our plot, MPD and UniFrac distance are closely, positively related. **Answer 4c:** MPD may vary less because it averages phylogenetic distances across all species pairs, which smooths out differences between communities. UniFrac is sensitive to unique lineages, so even small changes in species composition can cause much larger shifts in its values.

B. Visualizing Phylogenetic Beta-Diversity

Now that we have our phylogenetically based community resemblance matrix, we can visualize phylogenetic diversity among samples using the same techniques that we used in the β -diversity module from earlier in the course.

In the R code chunk below, do the following:

- perform a PCoA based on the UniFrac distances, and
- calculate the explained variation for the first three PCoA axes.

```
pond.pcoa <- cmdscale(dist.uf, eig = T, k = 3)

explainvar1 <- round(pond.pcoa$eig[1] / sum(pond.pcoa$eig), 3) * 100
explainvar2 <- round(pond.pcoa$eig[2] / sum(pond.pcoa$eig), 3) * 100
explainvar3 <- round(pond.pcoa$eig[3] / sum(pond.pcoa$eig), 3) * 100
sum.eig <- sum(explainvar1, explainvar2, explainvar3)
sum.eig
```

```
## [1] 20.9
```

Now that we have calculated our PCoA, we can plot the results.

In the R code chunk below, do the following:

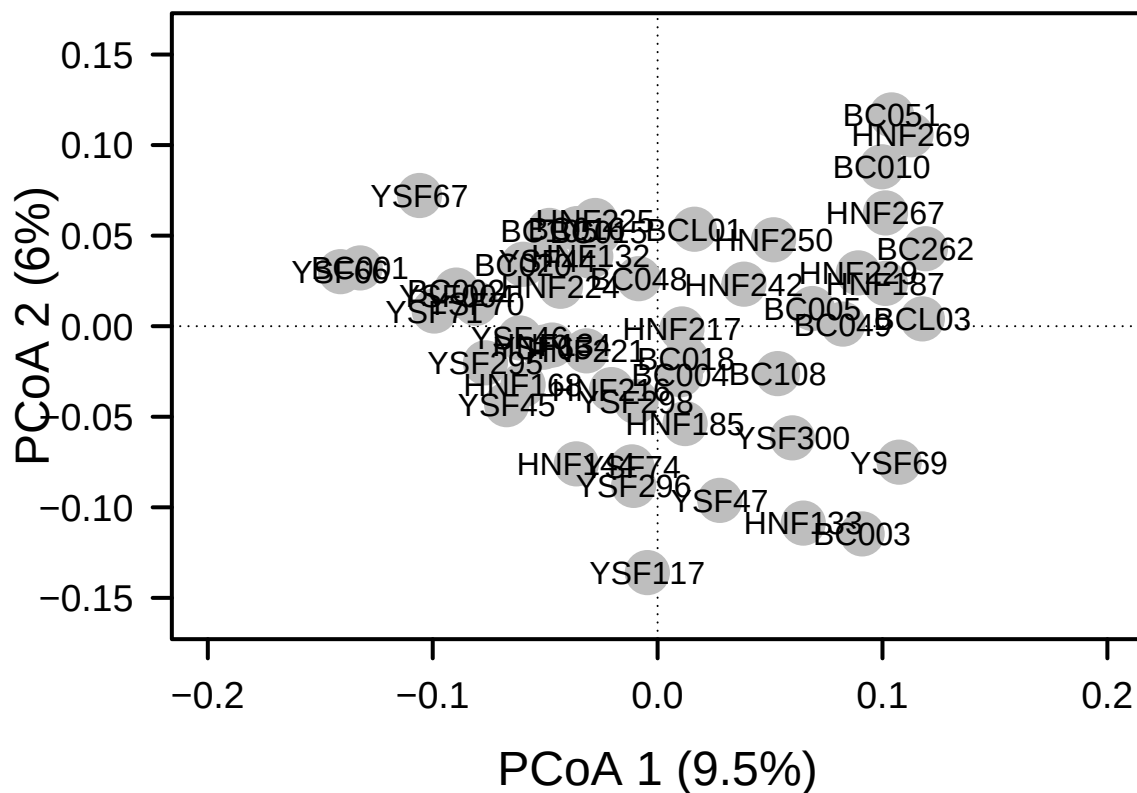
1. plot the PCoA results using either the R base package or the `ggplot` package,
2. include the appropriate axes,
3. add and label the points, and
4. customize the plot.

```
# Define Plot Parameters
par(mar = c(5, 5, 1, 2) + 0.1)

# Initiate Plot
plot(pond.pcoa$points[,1], pond.pcoa$points[,2],
     xlim = c(-0.2, 0.2), ylim = c(-.16, 0.16),
     xlab = paste("PCoA 1 (", explainvar1, "%)", sep = ""),
     ylab = paste("PCoA 2 (", explainvar2, "%)", sep = ""),
     pch = 16, cex = 2.0, type = "n", cex.lab = 1.5, cex.axis = 1.2, axes = FALSE)

# Add Axes
axis(side = 1, labels = T, lwd.ticks = 2, cex.axis = 1.2, las = 1)
axis(side = 2, labels = T, lwd.ticks = 2, cex.axis = 1.2, las = 1)
abline(h = 0, v = 0, lty = 3)
box(lwd = 2)

# Add Points & Labels
points(pond.pcoa$points[,1], pond.pcoa$points[,2],
       pch = 19, cex = 3, bg = "gray", col = "gray")
text(pond.pcoa$points[,1], pond.pcoa$points[,2],
     labels = row.names(pond.pcoa$points))
```



In the following R code chunk: 1. perform another PCoA on taxonomic data using an appropriate measure of dissimilarity, and 2. calculate the explained variation on the first three PCoA axes.

```
# Convert to presence-absence
comm.pa <- decostand(comm, method = "pa")

# Calculate Jaccard dissimilarity
tax.dist <- vegdist(comm.pa, method = "jaccard")

# Perform PCoA
tax.pcoa <- cmdscale(tax.dist, eig = TRUE, k = nrow(comm.pa) - 1)

# Extract eigenvalues
eigvals <- tax.pcoa$eig

# Calculate percent variation explained
explained <- eigvals / sum(eigvals) * 100

# Variation explained by first three axes
explained[1:3]
```

```
## [1] 8.831095 5.420578 4.532293
```

```
# 8.831095 5.420578 4.532293
```

```
explainvar1.t <- round(tax.pcoa$eig[1] / sum(tax.pcoa$eig), 3) * 100
```

```

explainvar2.t <- round(tax.pcoa$eig[2] / sum(tax.pcoa$eig), 3) * 100
explainvar3.t <- round(tax.pcoa$eig[3] / sum(tax.pcoa$eig), 3) * 100
sum.eig.t <- sum(explainvar1, explainvar2, explainvar3)
sum.eig.t

```

```
## [1] 20.9
```

Question 5: Using a combination of visualization tools and percent variation explained, how does the phylogenetically based ordination compare or contrast with the taxonomic ordination? What does this tell you about the importance of phylogenetic information in this system?

Answer 5: There is more clustering in the phylogenetic ordination than the taxonomic ordination. The same amount of variance was explained by the different ordination methods. This shows that it is important to consider the role of phylogenetic diversity in community composition because the taxonomic ordination and NTI values suggest species that are most abundant in the community tend to be closely related to each other, which could be overlooked with the taxonomic ordination alone.

C. Hypothesis Testing

i. Categorical Approach

In the R code chunk below, do the following:

1. test the hypothesis that watershed has an effect on the phylogenetic diversity of bacterial communities.

```

# Define Environmental Category
watershed <- env$Location

# Run PERMANOVA with `adonis()` Function {vegan}
phylo.adonis <- adonis2(dist.uf ~ watershed, permutations = 999)

# We can compare to PERMANOVA results based on taxonomy
tax.adonis <- adonis2(
  vegdist(
    decostand(comm, method = "log"), # create a distance matrix on
    method = "bray") ~ watershed,   # log-transformed relative abundances
    permutations = 999)              # using Bray-Curtis dissimilarity metric

```

ii. Continuous Approach

In the R code chunk below, do the following: 1. from the environmental data matrix, subset the variables related to physical and chemical properties of the ponds, and 2. calculate environmental distance between ponds based on the Euclidean distance between sites in the environmental data matrix (after transforming and centering using `scale()`).

```

# Define environmental variables
envs <- env[, 5:19]

# Remove redundant variables
envs <- envs[, -which(names(envs) %in% c("TDS", "Salinity", "Cal_Volume"))]

# Create distance matrix for environmental variables
env.dist <- vegdist(scale(envs), method = "euclid")

```

In the R code chunk below, do the following:

1. conduct a Mantel test to evaluate whether or not UniFrac distance is correlated with environmental variation.

```
# Conduct Mantel Test (`vegan`)
mantel(dist.uf, env.dist)

##
## Mantel statistic based on Pearson's product-moment correlation
##
## Call:
## mantel(xdis = dist.uf, ydis = env.dist)
##
## Mantel statistic r: 0.1604
##      Significance: 0.055
##
## Upper quantiles of permutations (null model):
##   90%   95% 97.5%   99%
## 0.129 0.165 0.200 0.236
## Permutation: free
## Number of permutations: 999
```

Last, conduct a distance-based Redundancy Analysis (dbRDA).

In the R code chunk below, do the following:

1. conduct a dbRDA to test the hypothesis that environmental variation effects the phylogenetic diversity of bacterial communities,
2. use a permutation test to determine significance, and 3. plot the dbRDA results

```
# Conduct dbRDA (`vegan`)
ponds.dbrda <- vegan::dbrda(dist.uf ~ ., data = as.data.frame(scale(envs)))

# Permutation tests: axes and environmental variables
anova(ponds.dbrda, by = "axis")

## Permutation test for dbrda under reduced model
## Forward tests for axes
## Permutation: free
## Number of permutations: 999
##
## Model: vegan::dbrda(formula = dist.uf ~ Elevation + Diameter + Depth + ORP + Temp + SpC + DO + pH + C)
##      Df SumOfSqs      F Pr(>F)
## dbRDA1   1  0.10566 2.0152  0.454
## dbRDA2   1  0.09258 1.8111  0.590
## dbRDA3   1  0.07555 1.5148  0.942
## dbRDA4   1  0.06677 1.3714  0.981
## dbRDA5   1  0.05666 1.1915  0.999
## dbRDA6   1  0.05293 1.1389  0.999
## dbRDA7   1  0.04750 1.0453  1.000
## dbRDA8   1  0.03941 0.7517
## dbRDA9   1  0.03775 0.7201
## dbRDA10  1  0.03280 0.6256
## dbRDA11  1  0.02876 0.5485
```

```
## dbRDA12    1  0.02501 0.4770
## Residual 39  2.04482
```

```
ponds.fit <- envfit(ponds.dbrda, envs, perm = 999)
ponds.fit
```

```
##
## ***VECTORS
##
##          dbRDA1    dbRDA2      r2 Pr(>r)
## Elevation  0.77670  0.62986 0.0959  0.079 .
## Diameter  -0.27972 -0.96008 0.0541  0.262
## Depth      -0.63137  0.77548 0.1756  0.008 **
## ORP         0.41879 -0.90808 0.1437  0.037 *
## Temp       -0.98250  0.18628 0.1523  0.013 *
## SpC        -0.77101  0.63682 0.2087  0.005 **
## DO         -0.39318 -0.91946 0.0464  0.326
## pH         -0.96210 -0.27270 0.1756  0.012 *
## Color       0.06353  0.99798 0.0464  0.326
## chla      -0.60392 -0.79704 0.2626  0.006 **
## DOC         0.99847 -0.05526 0.0382  0.401
## DON        -0.91633  0.40042 0.0339  0.450
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## Permutation: free
## Number of permutations: 999
```

```
# Calculate explained variation
dbrda.explainvar1 <- round(ponds.dbrda$CCA$eig[1] /
                           sum(c(ponds.dbrda$CCA$eig, ponds.dbrda$CA$eig)), 3) * 100
dbrda.explainvar2 <- round(ponds.dbrda$CCA$eig[2] /
                           sum(c(ponds.dbrda$CCA$eig, ponds.dbrda$CA$eig)), 3) * 100

# Make dbRDA plot

# Extract scores from the dbRDA object
ponds_scores <- vegan::scores(ponds.dbrda, display = "sites")

# Define plot parameters
par(mar = c(5, 5, 4, 4) + 0.1)

# Initiate plot
plot(ponds_scores, xlim = c(-2, 2), ylim = c(-2, 2),
     xlab = paste("dbRDA 1 (", dbrda.explainvar1, "%)", sep = ""),
     ylab = paste("dbRDA 2 (", dbrda.explainvar2, "%)", sep = ""),
     pch = 16, cex = 2.0, type = "n", cex.lab = 1.5, cex.axis = 1.2, axes = FALSE)

# Add axes
axis(side = 1, labels = T, lwd.ticks = 2, cex.axis = 1.2, las = 1)
axis(side = 2, labels = T, lwd.ticks = 2, cex.axis = 1.2, las = 1)
abline(h = 0, v = 0, lty = 3)
box(lwd = 2)
```



```

# Extract site scores
wa_scores <- vegan::scores(ponds.dbrda, display = "sites")

# Add points and labels
points(wa_scores,
      pch = 19, cex = 3, col = "gray")

# Add labels
text(wa_scores,
     labels = rownames(wa_scores),
     cex = 0.5)

# Extract environmental vectors (biplot scores)
vectors <- vegan::scores(ponds.dbrda, display = "bp")

# Add environmental vectors to the plot
arrows(0, 0, vectors[,1] * 2, vectors[,2] * 2,
      lwd = 2, lty = 1, length = 0.2, col = "red")

# Add labels for the environmental vectors
text(vectors[,1] * 2, vectors[,2] * 2, pos = 3,
     labels = rownames(vectors))

# Add axes for the vectors
axis(side = 3, lwd.ticks = 2, cex.axis = 1.2, las = 1, col = "red", lwd = 2.2,
     at = pretty(range(vectors[, 1]) * 2),
     labels = pretty(range(vectors[, 1]) * 2))

axis(side = 4, lwd.ticks = 2, cex.axis = 1.2, las = 1, col = "red", lwd = 2.2,
     at = pretty(range(vectors[, 2]) * 2),
     labels = pretty(range(vectors[, 2]) * 2))

```

